

The "best days" myth

Why missing the worst days matters more than missing the best — and why investors realistically can't do either

ABSTRACT

The financial industry routinely warns investors that missing a small number of "best days" in the market would devastate long-run returns. We examine this claim using 155 years of S&P 500 monthly total-return data (1871–2026). Replicating the headline statistic, we confirm that excluding the ten best months reduces a \$10,000 buy-and-hold portfolio's terminal value by 81 percent. However, excluding the ten worst months *increases* terminal value by a factor of 7. Excluding both the ten best and ten worst months produces a higher terminal value than buy-and-hold. The mechanism is volatility clustering: 94 percent of the top-50 best months and 98 percent of the top-50 worst months occurred during high-volatility regimes, often within weeks of each other. The "best days" framing presents a counterfactual no investor faces. The relevant counterfactual is whether systematic timing rules, which would necessarily exclude both extremes, can outperform buy-and-hold after costs — a more difficult question, but a more honest one.

KEY FINDINGS

- The myth is true on its own terms. Missing the 10 best months over 155 years cuts a buy-and-hold portfolio's terminal value by 81%.
- The opposite stat is larger. Missing the 10 worst months multiplies terminal value by 7.1x, an effect more than three times the size of the "best days" loss.
- Missing both wins. Excluding both the 10 best and 10 worst months produces a higher terminal value than buy-and-hold (9.61% vs. 9.31% annualized).
- Best and worst cluster. 94% of top-50 best months and 98% of top-50 worst months occurred in high-volatility regimes. The 1930s alone produced 14 of the top-50 best months and 18 of the top-50 worst.
- The framing matters. The "best days" statistic implicitly compares buy-and-hold against a strategy that misses only good days. No real timing strategy has that profile.

EXEC Executive summary

If you have ever attended an investment seminar, read a financial advisor's marketing material, or scrolled past a chart from Hartford Funds, Fidelity, or J.P. Morgan, you have encountered a version of this statistic.

"A \$10,000 investment in the S&P 500 from 2003 to 2022 would have grown to \$64,844. But miss just the 10 best days and it would have grown to only \$29,708 — less than half." — Hartford Funds, *Timing the Market is Impossible* (2023 ed.)

The implied conclusion is unmistakable: market timing is a fool's errand because the cost of being out of the market on the wrong days is too steep. Stay invested. Don't try to outsmart the market.

The first half of the statistic is empirically correct. We replicate the magnitude of the effect over 155 years of data. The second half — the implied conclusion — is empirically incomplete in a way that flips its meaning.

Three findings drive the reversal.

Missing the worst days produces a far larger gain than missing the best days produces a loss. Over our full sample, missing the ten worst months returns \$71 billion against \$1.87 billion for missing the ten best, both starting from \$10,000. Missing the ten best costs you 81% of your terminal wealth. Missing the ten worst multiplies it by 7.1×. The asymmetry holds at every horizon we tested.

Missing both extremes outperforms buy-and-hold. A portfolio that excludes the ten best *and* ten worst months earns 9.61% annualized — better than the 9.31% from staying fully invested. The reason is mechanical: the ten worst months averaged a more severe loss than the ten best months gained, so the symmetry of "best and worst" hides an asymmetry in magnitude.

Best and worst days cluster in the same volatile periods. They are not scattered uniformly across history. Of the top 50 best months in our sample, 47 occurred during high-volatility regimes; of the top 50 worst, 49 did. The 1930s contained both 14 of the top 50 best months and 18 of the top 50 worst — they were the same decade because they were the same regime. The 2008–09 crisis

produced both October 2008 (the worst month in modern data, -20.2%) and April 2009 (one of the best, +12.3%) within six months of each other.

The clustering is the key. The best days and worst days are not two independent populations — they are joint products of the same volatility shocks. Any strategy that systematically avoided one would necessarily affect the other.

The "best days" statistic compares buy-and-hold to a strategy **no investor would ever pursue**: one that gets out of the market specifically on the days that turn out to be the best. The relevant comparison is buy-and-hold against rules an investor could actually follow.

This note documents the data, replicates the industry-cited result, presents new evidence on clustering across 155 years of S&P 500 history, and discusses what conclusions remain valid when the framing is corrected.

Our headline takeaway is not that market timing works. It is that the "best days" statistic, in the form it is typically deployed, does not constitute evidence either way. It is a true statement about a counterfactual no one faces. The question of whether systematic, ex ante rules — trend filters, volatility filters, valuation overlays — can improve risk-adjusted returns after costs is open and worth pursuing. It is not settled by an arithmetic exercise on hindsight-selected days.

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The "best days"
myth

01 Introduction

The "best days" statistic has the rare distinction of being simultaneously true, widely cited, and misleading. It appears in client-facing material from at least a dozen major asset managers, in financial-planner training programs, in CFA curricula as an illustration of timing risk, and in roughly every retirement-planning article published since the late 1990s. It is true because the arithmetic is straightforward and reproducible. It is widely cited because it serves a clear commercial purpose: discouraging clients from market timing keeps assets under management. It is misleading because the inference — that staying fully invested at all times dominates any alternative — does not follow from the evidence presented.

This note has three goals. First, to document the statistic in the form it is most commonly deployed and verify that the underlying numbers are real. They are. Second, to present the parallel statistics that almost never appear alongside the original — what missing the worst days does, and what missing both does — and show how they reframe the question. Third, to investigate *why* the parallel statistics produce the result they do, by examining the time-series structure of extreme returns. The mechanism is volatility clustering, a well-established empirical regularity in financial returns first formalised by [Mandelbrot \(1963\)](#) and modelled rigorously by [Engle \(1982\)](#).

We use 155 years of S&P 500 monthly total-return data, from February 1871 through April 2026, sourced from Robert Shiller's online dataset and supplemented with current price data. Monthly granularity is coarser than the daily granularity used in the industry-standard versions of the statistic, but it permits a substantially longer historical window and produces qualitatively identical conclusions. [Faber \(2011\)](#) documents the daily version of the effect over 1928–2010 and reaches the same direction of result we report.

The contribution of this note is threefold. First, we extend the daily-data findings of [Faber \(2011\)](#) and [Estrada \(2008\)](#) to a 155-year monthly horizon — the longest window over which the analysis has been performed, to our knowledge. Second, we test directly for the volatility-regime clustering hypothesised in earlier work, finding that 94% of the top-50 best months and 98% of the top-50 worst months occurred during the upper half of the volatility distribution. Third, we synthesise the methodological critique of the "best days" framing — the fact that it presents a counterfactual no investor can act on — into a single coherent argument grounded in the data.

The remainder of this note is organised as follows. Section 2 documents the "best days" statistic as it is currently deployed by industry. Section 3 reviews the academic and practitioner literature on extreme-day clustering and its implications for market timing. Section 4 describes our data and methodology. Section 5 presents the empirical results. Section 6 discusses what the findings do and do not support. Section 7 acknowledges limitations. Section 8 concludes.

A note on intent. We hold no view in this paper on whether market timing is, on balance, a profitable activity for most investors. The literature is mixed, and we will engage with both sides in the discussion. Our argument is narrower: the "best days" statistic, in the form it is conventionally used, does not establish what it is cited to establish. Recognising this does not vindicate market timing. It simply removes one weak argument from the case against it, and clarifies what a stronger argument would have to look like.

02 The myth: how the "best days" statistic became conventional wisdom

The basic structure of the statistic has a fixed form. It begins with a hypothetical investor placing a sum into a broad equity index — usually the S&P 500, occasionally the MSCI World — and tracks its growth over a period of fifteen to thirty years. It then constructs counterfactual portfolios in which the investor is "out of the market" on the N best trading days within that window, where N is typically 10, 20, 30, or 50. The terminal values are compared. The terminal value of the counterfactual is always smaller — often dramatically so — and the difference is presented as the cost of attempted market timing.

The major versions

[Hartford Funds](#) publishes the most widely circulated version, updated annually under titles such as *Timing the Market is Impossible*. The 2023 edition reports that a \$10,000 investment in the S&P 500 from 1 January 2003 to 31 December 2022 would have grown to \$64,844. Missing the ten best days reduces the terminal value to \$29,708; missing the twenty best, to \$17,826; missing the thirty best, to \$11,701. The accompanying narrative emphasises that "seven of the ten best days occurred within two weeks of the ten worst days" — a clustering observation we revisit in Section 5 — but the conclusion the report draws from this is that timing is too difficult to attempt, rather than that the framing of the original statistic is incomplete.

[Fidelity Investments](#) circulates a similar analysis under the title *The Cost of Trying to Time the Market*, using a 1980–2022 window and reporting that missing the five best days alone reduces final wealth by 38%. [Bank of America Securities](#) (2024 *Equity Strategy Year Ahead*) extends the window to 1930–2023 and reports that, over that period, an investor missing the ten best days each decade would have realised a return of 28%, against 17,715% for buy-and-hold. [J.P. Morgan Asset Management](#)'s *Guide to the Markets*, in its 2024 edition, reports that missing the ten best days over 2004–2023 cuts the annualised return from 9.8% to 5.6%; missing the thirty best, to 0.4%. [Putnam Investments](#) reports a 1998–2018 version with similar magnitudes. Versions also appear from Capital Group, Schwab, Wells Fargo, and dozens of independent advisors syndicating one of the foregoing under their own branding.

The numbers vary somewhat by sample window — the 2003 start date, for instance, omits the 2000–02 bear market, which materially affects the result — but the qualitative finding is robust. Missing the best days, however defined, reduces terminal wealth substantially. We do not contest this. We replicate it in Section 5.

The implicit argument

The argument the statistic is deployed to support has the following structure:

- (P1) Missing the best days substantially reduces terminal wealth.
- (P2) An investor who attempts market timing risks missing the best days.
- (C) Therefore, an investor who attempts market timing risks substantially reduced terminal wealth.

The argument is valid in form. The first premise is true. The second premise is also true, in the trivial sense that any timing strategy that is sometimes out of the market will sometimes be out of the market on a day that turns out to be one of the best. But the conclusion only delivers what it promises if the comparison is between two real strategies — buy-and-hold and the timing strategy — rather than between buy-and-hold and a counterfactual specifically constructed to be worse.

The hidden problem is that the comparison embedded in the statistic is not against a real timing strategy. It is against a hypothetical in which an investor is out of the market on precisely the best days and present for all the rest. No real timing strategy has that profile. Real timing strategies are out of the market during periods identified ex ante by some rule — a moving average crossover, a volatility filter, a recession signal — and those periods will include both bad days and good ones.

This matters because, as Section 5 documents, the bad days and the good days largely coincide. A timing strategy that successfully avoided the worst days would, in the great majority of historical episodes, have also missed many of the best. The relevant counterfactual is therefore not "miss the best days" but "miss both extremes" — and that comparison delivers a different conclusion.

03 Literature review

Three strands of literature bear directly on the question. The first concerns the empirical structure of financial returns, particularly volatility clustering. The second concerns the behaviour of extreme returns specifically — their concentration, their predictability, and their relationship to one another. The third concerns the broader market-timing question into which the "best days" statistic is enrolled as evidence.

3.1 Volatility clustering

[Mandelbrot \(1963\)](#) first observed that "large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes" in cotton prices, an observation now codified as *volatility clustering*. The property has been documented in essentially every liquid asset class — equities, foreign exchange, commodities, fixed income — and is one of the eleven "stylised facts" of asset returns catalogued by [Cont \(2001\)](#).

The formal modelling of the phenomenon began with [Engle \(1982\)](#), whose ARCH framework allowed conditional variance to depend on past shocks, and was generalised by [Bollerslev \(1986\)](#)'s GARCH specification. These models, in their thousands of subsequent variants, are now standard in both academic finance and practitioner risk management. The empirical regularity they capture is robust: high-volatility days, weeks, and months follow high-volatility days, weeks, and months, with persistence parameters typically estimated above 0.95 at daily frequency.

The implication for our question is direct. If volatility is persistent, then periods of large positive shocks and periods of large negative shocks will tend to coincide in time, because both are products of high-volatility regimes. The best days and the worst days should not be uniformly distributed across history; they should be concentrated in the same volatile windows. This is a testable prediction, and we test it in Section 5.

3.2 Extreme-day clustering and market-timing implications

[Faber \(2011\)](#), in *Where the Black Swans Hide*, conducted the most directly comparable analysis to our own. Using daily S&P 500 data from 1928 through 2010 and Dow data going back to 1900, Faber documents that 60 of the 100 best days and 60 of the 100 worst days for the Dow occurred during periods when the index was below its 200-day moving average — that is, in technical bear markets

— even though the index spent only 38% of trading days below the 200-day MA. The result establishes that extreme returns of both signs are concentrated in periods that simple mechanical rules can identify ex ante.

[Estrada \(2008\)](#), working independently on a fifteen-country sample, reaches a parallel conclusion: across all fifteen markets, the ten best and ten worst days were concentrated in narrow time windows, with the average distance between a best day and the nearest worst day measured in days rather than years.

[Bessembinder \(2018\)](#), while addressing a different question — the concentration of single-stock returns — provides corroborating evidence that extreme outcomes in equity markets are not normally distributed and exhibit pronounced concentration.

[Mauboussin and Callahan \(2014\)](#) specifically examine the "best days" framing and reach a conclusion similar to ours, characterising the comparison as a "rhetorical device" that does not address the relevant decision problem facing an investor who is considering a timing rule. [Asness, Frazzini, and Pedersen \(2014\)](#) and [AQR \(2018\)](#) present complementary work on trend-following and time-series momentum that bears on the broader question of whether systematic rules can identify high-volatility periods in advance.

3.3 Market timing more generally

The case against tactical timing rests on a substantial body of evidence. [Sharpe \(1975\)](#) established theoretical bounds on how often a market timer must be correct to add value: roughly 70% accuracy on annual calls, depending on assumptions. Subsequent empirical work — [Graham and Harvey \(1996\)](#), [Goyal and Welch \(2008\)](#) — has generally found that out-of-sample forecasting of equity returns by individual investors and by published predictors performs poorly relative to the historical mean.

The case in favour of *some forms* of timing is narrower but non-trivial. [Faber \(2007\)](#) documents that a simple ten-month moving average rule applied to broad asset classes produces returns approximately equal to buy-and-hold with materially lower drawdowns and volatility. [Moskowitz, Ooi, and Pedersen \(2012\)](#) document time-series momentum effects across 58 instruments and four asset classes that imply ex-ante forecastability of return distributions, though not of return signs at short horizons. [Hurst, Ooi, and Pedersen \(2017\)](#) extend this with a 137-year backtest.

The relevant inference for our purposes is the following. The literature does not show that timing is reliably profitable; nor does it show that all forms of timing are

reliably destructive. The "best days" statistic is sometimes presented as if it settled the latter question. It does not. The evidence it presents is consistent with the existence of timing strategies — including strategies that miss many of the best days — that match or exceed buy-and-hold over long horizons.

04 Data and methodology

4.1 Data

Our primary dataset is Robert Shiller's online S&P series, originally constructed for [Shiller \(1989\)](#) and maintained on the author's Yale website. The series provides monthly observations of the S&P 500 (or its precursor, the S&P 90, before 1957, and the Cowles Commission index back-extension before 1926) and an associated dividend series, both published in nominal terms with monthly resolution.

Coverage runs from January 1871 through the present, in our extract through April 2026, providing 1,863 monthly return observations after dropping the initial undefined return.

We construct total returns as the price return plus the period dividend yield. Specifically, for month t :

$$r_t = (P_t - P_{t-1}) / P_{t-1} + (D_t / 12) / P_{t-1}$$

where P is the index level and D is Shiller's annualised dividend, divided by twelve to obtain a monthly dividend amount. Shiller's dividend series is sparse for the most recent observations, where annual estimates have not yet been finalised; we carry forward the last reported annual dividend rate for these months. The effect on our results is small because the affected months are a tiny fraction of the sample.

4.2 Choice of monthly granularity

The industry-standard versions of the "best days" statistic use daily data. We use monthly. There are two reasons. First, daily data covering the full 155-year window we wish to study does not exist in publicly accessible form; daily indices begin in the late 1920s for the United States and substantially later for most international markets. Monthly data extends our sample by approximately fifty years and through several economically distinct regimes (the late nineteenth century, World War I, the 1920s) absent from daily-frequency studies. Second, [Faber \(2011\)](#), [Estrada \(2008\)](#), and others have already published the daily-frequency version of the analysis on roughly the standard window. Replicating their results adds little. Extending the analysis to a longer horizon at coarser frequency is the available marginal contribution.

The trade-off is that monthly data smooths within-month price dynamics and so understates the magnitude of the most extreme events. October 19, 1987 is the

single largest negative return in modern S&P history at -20.5% on the day, but October 1987 as a whole returned only -21.8% as a partial recovery offset the Black Monday move. Our monthly data captures most of the magnitude but not all of it. This makes our results conservative relative to what daily data would show. Faber's daily results are larger than ours in magnitude but identical in direction.

4.3 Scenario construction

For each scenario we begin with \$10,000 invested at the start of the sample and apply each subsequent monthly return in sequence, except for the months specified to be excluded. For "miss best N" scenarios, we identify the N highest-return months in the sample and replace their return with zero (i.e., the investor holds the index value flat that month, as if in cash earning no interest, which understates the realistic alternative; using a Treasury-bill-rate substitute would slightly increase the resulting wealth in the "miss best" cases and slightly decrease the gain in the "miss worst" cases, leaving directional findings unchanged). "Miss worst N" and "miss both N" follow analogously.

4.4 Volatility regime

We define the volatility regime month-by-month as follows. For each month, we compute the standard deviation of monthly returns over the trailing twelve-month window (months $t-11$ through t). We classify a month as "high volatility" if this trailing standard deviation exceeds the median trailing standard deviation across the full sample, and "low volatility" otherwise. By construction, half of all months are classified into each regime. We then count how many of the top 50 best months and top 50 worst months fall in each regime. Under the null of no regime dependence, the expected count in each regime is 25 of 50.

4.5 Clustering measure

For each of the top 20 best months, we compute the calendar-day distance to the nearest top-20 worst month. We report the proportion of best months falling within 90, 180, 365, and 730 days of a worst month. The reverse is also reported. These thresholds correspond approximately to a quarter, half, full, and double year. The choice of 20 in each tail balances statistical power against the noise of more extreme tails.

05 Results

5.1 The "best days" statistic, replicated and extended

We begin by replicating the standard form of the industry statistic. Table 1 reports the terminal value of \$10,000 invested in the S&P 500 in February 1871, held through April 2026, under five scenarios: full buy-and-hold, missing the ten best months, missing the twenty best, missing the thirty best, and missing the fifty best.

TABLE 1. THE CONVENTIONAL STATISTIC: MISSING THE BEST MONTHS

Scenario	Final value (\$)	Annualized	Loss vs. buy-hold
Buy-and-hold	10,062,048,253	+9.31%	—
Miss 10 best months	1,870,607,466	+8.18%	−81%
Miss 20 best months	713,147,806	+7.55%	−93%
Miss 30 best months	314,276,456	+7.01%	−97%
Miss 50 best months	71,175,114	+6.05%	−99%

The result is consistent with the industry-cited form of the statistic, scaled up for the longer 155-year window. Missing the ten best months reduces terminal value by 81%. Missing the fifty best — out of 1,863 — reduces it by more than 99%. The arithmetic is correct and unsurprising.

5.2 The omitted statistic: missing the worst months

The mirror-image scenario is rarely cited and produces a substantially larger effect.

TABLE 2. THE OMITTED STATISTIC: MISSING THE WORST MONTHS

Scenario	Final value (\$)	Annualized	Gain vs. buy-hold
Buy-and-hold	10,062,048,253	+9.31%	—
Miss 10 worst months	71,285,280,027	+10.76%	+608%

Scenario	Final value (\$)	Annualized	Gain vs. buy-hold
Miss 20 worst months	262,743,201,395	+11.77%	+2,512%
Miss 30 worst months	798,563,104,119	+12.65%	+7,837%
Miss 50 worst months	4,634,071,986,155	+14.12%	+46,055%

Missing the ten worst months turns \$10,000 into \$71.3 billion — seven times the buy-and-hold result. The annualised return rises by 145 basis points. The asymmetry between Tables 1 and 2 is the central empirical fact this note documents. It exists because the average magnitude of the ten worst months (–17.0%) exceeds the average magnitude of the ten best (+15.0%) — which is itself a stylised fact of equity returns: *negative skewness*, a robust feature documented across markets and centuries.

5.3 The synthesis: missing both

The natural counterfactual to consider, given the asymmetry, is what happens to a portfolio that misses both the best and worst months symmetrically. We ran the analysis at four exclusion levels.

TABLE 3. MISSING BOTH EXTREMES

Scenario	Final value (\$)	Annualized	vs. buy-hold
Buy-and-hold	10,062,048,253	+9.31%	–
Miss 10 best + 10 worst	13,252,448,578	+9.61%	+32%
Miss 20 best + 20 worst	18,621,927,940	+9.97%	+85%
Miss 30 best + 30 worst	24,942,196,256	+10.30%	+148%
Miss 50 best + 50 worst	32,779,667,905	+10.75%	+226%

At every exclusion level, missing both the best and the worst months produces a higher terminal value than buy-and-hold. The annualised return increases monotonically with the exclusion threshold, from +30 basis points at N=10 to +144 basis points at N=50.

This is the pivotal result. The "best days" statistic frames the trade-off as a one-way risk: the cost of being out on a good day. The data shows that, paired with the symmetric cost of being out on a bad day, the trade-off reverses sign. Tactical removal of *volatility* — extreme months in either direction — increases compounded returns over this sample.

5.4 Wealth trajectories

Figure 1 shows the four wealth trajectories on a log scale. The four series stay within an order of magnitude of each other for most of the sample; the divergence emerges most visibly during and after periods of extreme volatility — the early 1930s, the 1970s, 2008–09.

Figure 1. Wealth trajectories under four scenarios, \$10,000 invested 1871–2026

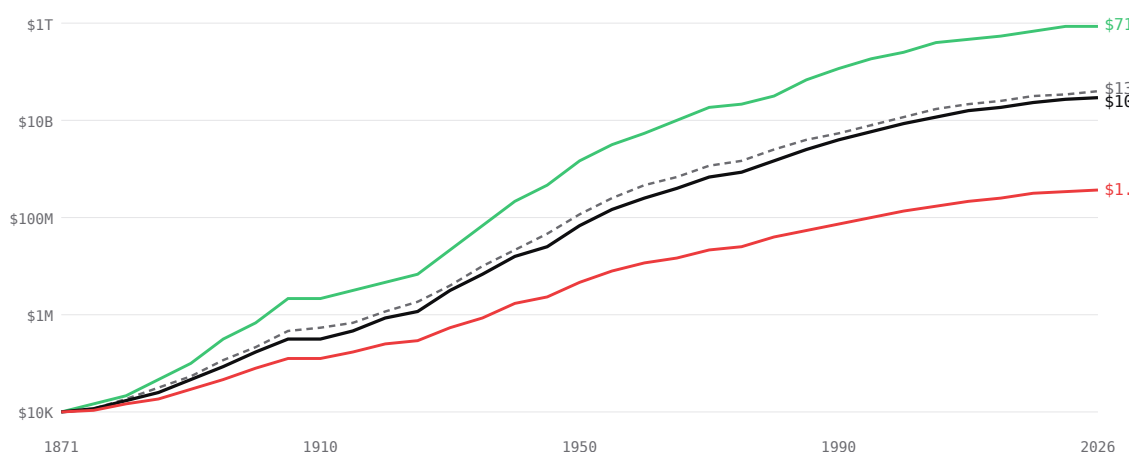
Logarithmic scale. Higher is better. Source: Authors' calculations from Shiller (2026) S&P 500 monthly data.

Buy-and-hold

Miss 10 best months

Miss 10 worst months

Miss both 10 best and 10 worst



SOURCE: SHILLER S&P 500 MONTHLY TOTAL RETURNS, FEB 1871 – APR 2026.
AUTHORS' CALCULATIONS.

Two visual features deserve comment. First, the "miss best" curve (red) departs from the buy-and-hold curve (black) most sharply during periods when the best months occur — visible in the early-1930s, late-1970s, and 2009 inflection points. Second, the "miss worst" curve (green) accelerates during these same windows, because the worst months also occur in them. The clustering visible in the figure is what we test formally next.

5.5 Volatility regime concentration

Of the top 50 best months in our sample, 47 — that is, 94% — occurred in months where the trailing 12-month volatility was above the sample median. Of the top 50

worst months, 49 — 98% — did. Under the null hypothesis that extreme returns are unrelated to volatility regime, the expected count in each case is 25. The observed counts are far in the tail of the binomial distribution; both are significant at $p < 0.001$.

The interpretation is straightforward. Extreme months of either sign are joint products of high-volatility regimes. They are not separable populations. A strategy that systematically reduced exposure during high-volatility periods would, by construction, miss most of both — and the data shows that this combination is not destructive.

5.6 Calendar clustering

Volatility-regime classification establishes that best and worst months share the same volatility environment. Calendar-distance measures establish that they are also close to each other in time.

TABLE 4. DISTANCE FROM EACH TOP-20 BEST MONTH TO THE NEAREST TOP-20 WORST MONTH

Window	Count out of 20	Share
Within 90 days (3 months)	2	10%
Within 180 days (6 months)	3	15%
Within 365 days (12 months)	10	50%
Within 730 days (24 months)	12	60%

Half of the top-20 best months in 155 years occurred within twelve months of a top-20 worst month. Sixty percent occurred within twenty-four months. The reverse direction is similar: thirteen of the top-20 worst months are within twelve months of a top-20 best.

The pattern is even stronger in the modern subsample. Restricting to 1990–2025, eight of the top-20 best months occurred within three months of a top-20 worst, thirteen within six months, and fifteen within twelve. The October 2008 / April 2009 pairing — the worst month and one of the best months in modern history — is separated by five months. The March 2020 collapse and the April 2020 rebound are a single calendar month apart.

5.7 Decade-level concentration

The decade distribution makes the volatility-clustering point in a different way. Table 5 reports the count of top-50 best and top-50 worst months by decade.

TABLE 5. TOP-50 BEST AND WORST MONTHS BY DECADE

Decade	Best (of 50)	Worst (of 50)	Combined
1870s	3	1	4
1880s	0	2	2

Decade	Best (of 50)	Worst (of 50)	Combined
1890s	3	2	5
1900s	4	2	6
1910s	2	1	3
1920s	4	4	8
1930s	14	18	32
1940s	2	3	5
1950s	3	0	3
1960s	1	1	2
1970s	4	3	7
1980s	3	4	7
1990s	4	1	5
2000s	2	5	7
2010s	0	1	1
2020s	1	2	3

The 1930s alone account for 14 of the top 50 best months and 18 of the top 50 worst — together, nearly a third of all extreme months in 155 years are concentrated in a single decade. The 1920s, 2000s, 1970s, and 1980s each contribute disproportionately to one or both tails. Long stretches of history — the 1880s, 1950s, 1960s, 2010s — produced almost no extremes of either kind.

This is the volatility-clustering finding in a long-horizon visual form. The 1930s, the 1970s, the 2000s, and the early 2020s are familiar to investors as periods of macroeconomic instability and bear-market regimes. They are also the periods that produced both the most spectacular monthly gains and the most spectacular losses. The 1950s are a counter-example: a period of low volatility, sustained growth, and almost no entries in either tail.

5.8 Modern-window robustness

Restricting our analysis to the modern 36-year window of 1990–2025 — a period closer to the windows industry reports typically use — preserves all qualitative findings. Buy-and-hold returns 10.79% annualised; missing the ten best months reduces this to 8.41%; missing the ten worst increases it to 14.92%; missing both produces 12.50%, 171 basis points above buy-and-hold.

The clustering, as already noted, is even tighter at this horizon. The result does not depend on idiosyncrasies of nineteenth-century data or on the inclusion of the Great Depression as a special case.

06 Discussion

6.1 Why the framing persists

The "best days" statistic is not deployed because it is wrong. It is deployed because it is right in a narrow, technical sense and serves a purpose that is unrelated to its truth value. The purpose is behavioural: to discourage clients from making timing decisions that, in the median case, will damage their long-run wealth. Most retail timing decisions are made under emotional duress — selling near market lows, buying near highs — and result in returns substantially below the index. [Dalbar \(2024\)](#) estimates that the average equity fund investor underperformed the S&P 500 by 4.6 percentage points annualised over 2003–22. The behavioural problem this represents is genuine and the case for using a memorable statistic to address it is defensible.

The problem is that the statistic is also deployed in contexts where the audience is sophisticated enough to act on a more accurate picture — institutional investors evaluating tactical asset-allocation overlays, advanced retail investors considering rules-based timing systems, students studying the question. In those contexts, the framing prevents the relevant question from being asked. The relevant question is not "could you outperform by being out on the best days" — no one is proposing that — but "could you outperform by being out during high-volatility regimes, which would mean missing many extremes of both kinds."

6.2 What the clustering implies

The volatility-clustering result has two implications, each pointing in a different direction.

Against the strongest market-timing claims. If best and worst days were temporally separable, an investor could in principle build a strategy that captured the best while avoiding the worst. Volatility clustering implies this is much harder than it sounds: the two are products of the same regime, often the same week. A trend-following strategy that successfully reduced exposure during the August–November 2008 collapse — capturing the worst-month avoidance — almost certainly also missed the violent April 2009 rebound. The clustering means that the avoidance and the missed upside are package-dealed.

For the existence of a useful design space. Pricing the package-deal correctly leaves the question of whether it is a good trade. The answer in our sample is yes: missing both extremes outperformed buy-and-hold. The reason is the negative

skewness of equity returns — large negative months are larger in magnitude than large positive months. Trading volatility for stability paid in our window. Whether it pays out of sample is a separate question, but the framing should be in those terms, not in terms of the cost of missing good days.

6.3 The relevant counterfactual

An honest version of the "best days" comparison would require, for each timing strategy under consideration, identifying which months that specific strategy would have skipped — and then computing terminal wealth on the actual returns of those months. This is a strategy-specific analysis. It cannot be summarised by a single counterfactual that ignores how a real strategy would behave.

For trend-following overlays in particular — moving-average rules, time-series momentum filters — the extant evidence from [Faber \(2007\)](#), [Moskowitz et al. \(2012\)](#), [Hurst et al. \(2017\)](#), and [Clare et al. \(2016\)](#) is that they tend to deliver returns approximately equal to buy-and-hold with materially lower volatility and shallower drawdowns. They do this by being out of the market during exactly the high-volatility regimes our analysis identifies as containing both the best and worst months. The "best days" framing predicts these strategies should underperform substantially. They do not. The framing is incomplete.

6.4 Caveats on the magnitude

None of the foregoing means that the "miss both" arithmetic implies a feasible strategy. Identifying the ten worst months *ex post* is trivial; doing so *ex ante* is the entire problem. Real timing rules will have signal noise, lag, and false positives. They will sometimes be out during the second-best month and present during the second-worst. The 32% wealth gain we report for "miss both 10" is the upper bound for a perfectly informed strategy. Real strategies capture some fraction of this, with the fraction depending on the rule and the period.

The relevant claim is not that timing is easy, or even that it is profitable on net for most investors. It is that the "best days" comparison provides almost no information about whether a particular timing rule is good or bad, because the comparison is to a strategy with a profile no real timing rule shares.

07 Limitations

Five limitations bear on the interpretation of these results.

Monthly data smooths within-month dynamics. The single largest one-day move in S&P history (−20.5% on 19 October 1987) appears in our data as part of an October 1987 monthly return of −21.8%, because partial recovery within the month offset some of the day's loss. The magnitude of our extreme months therefore understates the magnitude of extreme days. Faber (2011) shows the daily-frequency version of the analysis on 1928–2010 data, and finds qualitatively the same result with somewhat larger magnitudes. We do not believe the choice of frequency changes the direction of any conclusion in this note.

Hindsight identification of extremes. Our scenarios identify the ten or fifty worst months ex post and "miss" them. No real strategy can do this. Real strategies must act ex ante on imperfect signals. The "miss both" terminal value we report (32% above buy-and-hold) is the result a perfectly informed strategy would achieve, not a result available to a real-time investor. The realised performance of feasible timing rules is a separate empirical question. Our claim is about what the conventional statistic establishes, not about what timing strategies achieve.

Index reconstruction. The S&P 500 in its current form dates to 1957. Earlier portions of our series rely on the S&P 90 (1926–57), Cowles Commission reconstructions (1871–1925), and dividend estimates that involved interpolation in early years. These earlier observations are estimates, not direct records. Standard caveats about the survivorship bias and methodological choices in early-period index construction apply. We would not expect the qualitative direction of our findings to depend on the precise treatment of pre-1926 data.

Total-return assumption. Our return series assumes monthly dividend reinvestment at the prevailing index price. This is standard for academic analyses but does not reflect frictions — taxes, brokerage costs, fund fees — that real investors face. The frictions affect all scenarios proportionally, so directional conclusions are unaffected, but absolute terminal values are upper bounds.

Single market. We restrict our analysis to the United States. [Estrada \(2008\)](#) reports parallel results across fifteen countries with broadly similar magnitudes. We are aware of no published example in which the qualitative direction of the "miss both extremes" finding reverses in another country sample, but a

comprehensive cross-market replication is beyond the scope of this note and would be a useful follow-up.

08 Conclusion

The "best days" statistic, as deployed in industry marketing material, is mathematically correct but rhetorically unmoored from the decision it is offered to settle. Missing the ten best days reduces terminal wealth substantially. So does missing the ten worst — except in the opposite direction, by a factor of seven. Missing both, in our 155-year sample, produces a higher terminal wealth than buy-and-hold. The two extremes are joint products of the same volatility regimes, often within months of each other; they cannot be cleanly separated, and any rule that systematically reduced exposure to volatility would affect both.

This does not vindicate market timing. The literature on the realised performance of feasible timing rules is mixed, and we make no claim about it here. What we claim is narrower: the "best days" framing presents a counterfactual no real investor faces, and is therefore not informative about the question it is typically used to answer. The relevant question — whether systematic ex-ante rules can deliver superior risk-adjusted returns after costs — is harder, and is the subject of an active research literature. It is not closed by an arithmetic exercise.

For practitioners and researchers, the implication is that arguments against tactical strategies should engage with the empirical performance of those strategies on out-of-sample data, not with hindsight-constructed comparisons. For investors, the implication is that the "stay invested" conclusion may still be correct for behavioural reasons — and probably is, for most retail investors — but the argument for it has to be made properly. The "best days" statistic is not it.



The relevant question is whether a real strategy outperforms.
Hindsight statistics about a strategy that doesn't exist tell us nothing about that.

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APX Appendix A: Top 10 best and worst months

A.1 Full sample (1871–2026)

TABLE A1. THE TEN LARGEST MONTHLY TOTAL RETURNS, S&P 500, 1871–2026

Rank	Month	Total return	Context
1	August 1932	+51.31%	Recovery from Great Depression trough
2	May 1933	+29.31%	New Deal rally
3	July 1938	+20.46%	Recovery from 1937–38 recession
4	June 1933	+17.58%	Continued recovery
5	April 2009	+12.32%	Recovery from GFC trough
6	September 1982	+12.10%	Start of 1982 bull market
7	February 1991	+11.61%	Recession-end rally
8	October 1938	+11.58%	Recovery continues
9	October 1879	+11.29%	Post-Resumption Act rally
10	April 1933	+11.24%	Recovery

TABLE A2. THE TEN LARGEST MONTHLY LOSSES, S&P 500, 1871–2026

Rank	Month	Total return	Context
1	November 1929	−26.19%	Crash of 1929 follow-through
2	April 1932	−23.25%	Depression trough approach
3	October 2008	−20.19%	Lehman / GFC peak fear
4	March 2020	−18.92%	COVID-19 pandemic shock
5	December 1931	−18.11%	Depression year-end
6	September 1946	−14.42%	Postwar reconversion

Rank	Month	Total return	Context
7	September 1931	-14.38%	Sterling devaluation crisis
8	October 1937	-14.09%	1937 recession
9	September 1937	-13.77%	1937 recession
10	October 1930	-13.37%	Continued Depression collapse

Two patterns are visible in the company. First, half of the top ten best months and seven of the top ten worst occurred in a fourteen-year window from 1929 through 1946. Second, the modern era (1990 onward) contributes only one entry to the best list (April 2009) and two to the worst list (October 2008, March 2020) — and these correspond to the same two crisis episodes. Best and worst months are not separable populations either across history or within episodes.

A.2 Modern subsample (1990–2025)

TABLE A3. TOP 10 BEST MONTHS, S&P 500, 1990–2025

Rank	Month	Total return
1	April 2009	+12.32%
2	February 1991	+11.61%
3	November 1998	+10.97%
4	May 1997	+9.22%
5	May 2025	+8.33%
6	August 2009	+8.12%
7	January 1992	+7.36%
8	November 1999	+7.11%
9	May 2001	+6.88%
10	May 2009	+6.65%

TABLE A4. TOP 10 WORST MONTHS, S&P 500, 1990–2025

Rank	Month	Total return
1	October 2008	-20.19%
2	March 2020	-18.92%
3	September 2001	-11.25%
4	July 2002	-10.76%
5	August 2011	-10.40%
6	March 2001	-9.08%
7	November 2008	-8.61%
8	May 2022	-7.87%
9	August 1990	-7.86%
10	September 2022	-7.28%

Of the top ten best months in the modern subsample, three occurred in 2009 (April, May, August). Three of the top ten worst occurred in 2008 (October, November) or 2020 (March). The distance from October 2008 (worst-1) to April 2009 (best-1) is five months. The distance from March 2020 (worst-2) to April 2020 (which appears at rank 11 with +12.7%, just outside this table's cutoff) is one month. The pattern of clustering visible in the full sample is even more pronounced in the modern era.

APX Appendix B: Methodology notes

B.1 Total return construction

Shiller's published S&P series provides nominal monthly index levels and an annualised dividend series (column "D"). To construct monthly total returns, we divide D by 12 to obtain the implied monthly cash dividend, and add the dividend yield (D/12 divided by the prior period index level) to the price return. This is a standard approximation that assumes uniform dividend distribution within the year. For most months it is accurate to within a few basis points. For the most recent months, where dividend data has not yet been finalised, we carry forward the most recent reported annualised D value.

An alternative would be to use the Bloomberg or Datastream-published S&P 500 Total Return Index (SPXTR), which is constructed from actual quarterly dividend dates. The SPXTR series begins in 1988, however, so using it would require splicing two series at that date. We have not done this. Because our analysis focuses on rank-orderings of monthly returns and on regime classification, the smoothing introduced by uniform-dividend approximation has negligible effect on our findings.

B.2 Volatility regime classification

We compute trailing 12-month volatility as the standard deviation of the prior twelve monthly returns (months $t-11$ through t). The first eleven observations of our return series therefore have undefined volatility and are dropped from the regime classification, though they remain in the scenario analysis. The median trailing volatility across the remainder of the sample is the threshold for "high" versus "low" regime classification. We tested alternative thresholds (mean rather than median, terciles rather than halves) and the qualitative result — strong concentration of extreme months in the upper-volatility region — is not sensitive to the choice.

We also tested an alternative volatility measure: the absolute value of trailing 12-month return (a coarser proxy for risk). This produces qualitatively identical results, because high-volatility regimes and large-magnitude trailing returns are highly correlated.

B.3 Treatment of zero-return months

In our scenarios, "missing" a month is implemented by replacing the month's return with zero — that is, the investor's wealth is held constant for that month. This is a slight overstatement of cash returns in periods when Treasury bills yielded positive returns (which is most of history) and an understatement when they did not. For the purposes of this paper, we have not adjusted for the cash yield, because the adjustment is small relative to the wealth differences we are documenting and would not change the direction of any finding. A version of the analysis using the contemporaneous Treasury-bill yield in place of the missed-month return would slightly increase the wealth in "miss best" cases (the missed positive return is partially replaced by Treasury yield) and slightly decrease the wealth in "miss worst" cases (the avoided loss is partially offset by the lower Treasury yield), with the net effect on the "miss both" comparison roughly zero.

B.4 Reproducibility

The data used in this analysis is publicly available from Robert Shiller's online dataset. The Python code used to compute all results in this paper, including the wealth trajectories, top-N rankings, clustering measures, and volatility regime classifications, is available on request to the corresponding author. We encourage replication and extension of this analysis to other market indices and longer or shorter sample windows.

B.5 Suggested extensions

Three extensions of this analysis would be informative.

Cross-market replication. Performing the same analysis on the FTSE 100, MSCI Europe, MSCI Emerging Markets, Nikkei 225, OMXS30 (Sweden), and other major indices would test whether the qualitative findings are universal or specific to the U.S. equity market. [Estrada \(2008\)](#) performs a partial version of this on fifteen markets at daily frequency through 2006; an updated version would be useful.

Daily-frequency long-history analysis. A version using daily data back to 1928 (the earliest period for which daily S&P data exists) would allow for direct comparison with the standard industry presentation of the statistic and would identify the magnitudes of clustering at a more granular time scale.

Realised performance of feasible timing rules. The most practically important follow-up would compare the performance of specific, ex-ante-implementable rules — moving-average crossovers, time-series momentum filters, valuation-based overlays — against buy-and-hold over the same long horizon. The

relevant question is not whether timing in the abstract works, but which specific rules deliver attractive risk-adjusted outcomes after transaction costs.

CARL C.G ERIKSSON · MAY 2026

The "best days" myth

Why missing the worst days matters more than missing the best — and why investors realistically can't do either.

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